9.1.1 b/d

(b)

Since we have 26 Capital letters and 9 digits. thus for each character we have 26+9=35 choices

Since we are given that the license plate starts with a digit. Thus for the first character we have only 9 choices. But for rest of the characters, we have 35 choices

$9 \times 35 \times 35 \times 35 \times 35 \times 35 \times 35 = 9 \times 35^6$.

(d) For the first 3 we can choose any from 28 and for last 4 we can choose any From?

$28 \times 28 \times 28 \times 9 \times 9 \times 9 \times 9$

TOTAL license plate = $28^3 \times 9^4$.

9.2.2 b

the objective is to determine the number of passwords of string length 7, 8 or 9 in which characters can be digits, letter, or special character

1st char	2nd char	3rd char	4th char	5th char	6th char
40 choices	40 choices	40 choices	40 choices	40 choices	40 choices

Total numbers of password of string length 7 in which characters can be digits, letters, or special characte
$= 40^7$

1st char = 40 choices
2nd char = 40 choices
3rd char = 40 choices
4th char = 40 choices
5th char = 40 choices
6th char = 40 choices
7th char = 40 choices
8th char = 40 choices

Total numbers of password of string length 8 in which characters can be digits, letters, or special characte
$= 40^8$

1st Char = 40 Choices

2nd Char = 40 Choices
3rd Char = 40 Choices
4^{Th} Char = 40 Choices
5^{Th} Char = 40 Choices
6^{Th} Char = 40 Choices
7^{Th} Char = 40 Choices
8^{Th} Char = 40 Choices
9^{Th} Char = 40 Choices

Total numbers of password of String length 9 in which characters can be digits, letters, or Special character $= 40^9$

therefore, the Total number of password of String length 7, 8 or 9 in which Characters can be digits letters, or Special Characters $= 40^7 + 40^8 + 40^9$.

9.1.3 b/d

(b) Except Friday, each day Fiona has 10 options to Choose from.

On Friday she has 3 options to choose from.
So, number of different ways of Choosing her lunches for the week $= 10 \times 10 \times 10 \times 10 \times 3 = 30{,}000$.

(d) Each day Fiona has 10 meal options and 2 drinks options to choose from.
So, number of different ways of choosing her lunches for the week $= (10\times2)\times(10\times2)\times(10\times2)\times(10\times2)\times(10\times2) = 32{,}000$pc

9.1.4 b
$4^{98} \times 4$ ways

1. Both pages gets assigned to the first printer
2. Front page gets assigned to the firstprinter and back gets assigned to the second printer.

3. Back page gets assigned to the first printer and front page gets assigned to the second printer.

4. Both pages gets assigned to the second printer

9.2.2 b

$|P_6| = |B^3|$

Here B^3 hence means 3 places, each place has 2 options.

So $|B^3| = 2^3 = 8$

$\therefore |P_6| = 8.$

9.2.3 b

we know $|E_{10}| = |B^9| = 2^9$, since for each of the nine positions in any string B^9 we can independently choose one of two values for the bit.
therefore $|E_{10}| = 2^9 = 512.$

9.2.4 b
number of strings balanced $= 3^5 = 243$

9.2.5 b

the output will be the same

(Eli, Ike, Don, Hal, Jan, Abe, Ben, Fran, Gene, Cam)

9.3.1 b

1st → 36 ways
2nd → 39 ways

3rd → 38 WAYS
4th → 37 WAYS
5th → 36 WAYS
6th → 35 WAYS

the TOTAL number of passwords with string of length 6 is equal to $36 \times 39 \times 38 \times 37 \times 36 \times 35 = 2487270240$

9.33 b

Number of license plates possible if no digit appears more than once

$$D \times L \times L \times L \times (D-1) \times (D-2) - 1$$

$10 \times 26 \times 26 \times 26 \times 9 \times 8 =$
$= 26^4 \times 10 \times 9 \times 8$
$= 329022720$

9.4.1 b/d

(b) To find the number of strings of length 8 consecutive a's

a	a	a	a	a	a	a	a	8 Consecutive a's with another a
a	a	a	a	a	a	a	b	8 Consecutive a's with followed by b

a	a	a	a	a	a	a	c	8 consecutive a's with followed by c
b	a	a	a	a	a	a	a	b followed by 8 consecutive a's
c	a	a	a	a	a	a	a	c followed by 8 consecutive a's

(d)

Case 1 ⇒ the first character is same as last
the first character can be filled by 3 ways
while the remaining characters can be filled by 3^7 ways.
Number of strings in which first character is same as last = 6561

Case 2 ⇒ the last character is a
In this case, the first character cannot be a else it will be same as case 1.
So, for the first character can be filled by 2 ways and the remaing 7 characters by 3^7 ways. therefore, the total number of strings with last character as a is = $2 \times 3^7 \times 1 = 4374$.

Case 3 ⇒ the first character is a.
In this case, the last character cannot be a else it will be same as case 1.
So, for the last character can be filled by 2 ways. and the remaining 7 characters by 3^7 ways. there-

fore, the TOTAL number of String with First character as a is = $2 \times 3^7 \times 1 = 4374$

TOTAL WAYS = 6561 + 4374 + 4374
= 15309

9.4.2b

We have TOTAL 2^{10}. STrings
Now we need to Choose those which there is at least one 1 is present and one 0 is present. there is only two case WithouTany 1 or 0 preseet. (0000000000) or (1111111111).

So TOTAL Cases are $2^{10} - 2$.

9.4.3b

Let assume P = number of Students in math 2A

Q = numbers of students in math 2B
R = number of Students in math 2C
then $n(P) = 28$, $n(Q) = 28$, $n(R) = 28$.

$n(P \cap Q \cap R) = 3$ $n(P \cap Q) = 11$ $n(P \cap R) = 9$ $n(Q \cap R)$ 10

then

Numbers of Students in group 3 = $n(P \cup Q \cup R) =$

$28 + 28 + 25 - 11 - 9 - 10 + 3 = 54$

The group 3 on B has 54 students.

9.4.4 b

92 integers.

9.5.1 b

We have 2^7 choices for the first elements Image $f(a_{000000}) \; 2^7 - 1$ choices for the second element Image $f(a_{000001})$ and so on.

thus TOTAL number of choices is

$2^7 \times (2^7 - 1) \times (2^7 - 2) \cdots 1$

TOTAL is equal

$(27)! = (128)!$

9.5.2 b

Total 60460 different phone numbers.

9.5.3 b

9 Choices
$= 9 \times 9! = 3265920$

9.6.1
$[12, 3, 12, 4, 19]$ is not a S-permutation as a entries of S-permutation are district.

9.6.2 b/d

(b) $\{a, b, c, d\}$

(d) $\binom{7}{3} + \binom{7}{4} = \frac{7!}{3!(7-3)!} + \frac{7!}{4!(7-4)!} = 2 \times \frac{7 \times 6 \times 5 \times 4!}{4! \times 3!} =$
70

9.6.3 b/d

(b) If the string starts with 001. First three

bits are Fixed and hence each of the remaining 7 bits independently can be 0 or 1 in 2 ways. So, number of 10-bit strings which START with 001 is: $2^7 = 128$

(d) there are $2^2 = 4$ WAYS the FIRST TWO bits can form. Each of the remaining 6 bits independently can be 0 or 1 in 2 WAYS. So, required answer is: $4 \times 2^6 = 256$.

9.6.4

(b) So after filling 5 a's only 7 places remain, now all the remaining places have to be filled by b or c, so there are 2 choices for each places, that is $2^7 = 128$ WAYS, thus answer is $792 \times 128 = 101376$ WAYS.

9.6.6

$$\binom{\quad}{\quad} = \frac{37!}{2!(35)!} = 666$$

9.6.7

$${}_8C_3 \times {}_6C_2 = \frac{8!}{3!(8-3)!} \times \frac{6!}{2!(6-2)!}$$

$$\frac{8 \times 7 \times 6 \times 5!}{3 \times 2 \times 1 \times 5!} \times \frac{6 \times 5 \times 4!}{2 \times 1 \times 4!}$$

$8 \times 7 \times 3 \times 5 = 840$

9.6.8

${}^{13}C_2 \times {}^{39}C_3$

$$\frac{13!}{(2!)(13-2)!} \times \frac{39!}{(3!)(39-3)!}$$

$$\frac{13!}{(2!)(11)!} \times \frac{39!}{(3!)(36)!}$$

$78 \times 9139 : 712,842$

9.7.1

$P = (37,4)$
$P = (37-4)!$
$P = (33)!$
$P = 15\,85080$

9.7.2

${}^{30}P_5 =$

$= 30 \times 29 \times 28 \times 27 \times 26$

$= 17100720$ WAYS

9.7.3

$\frac{12!}{(12-5)!} =$

$\frac{12!}{7!} =$ 95040

9.7.4

$\frac{9!}{6!(9-6)!} =$

$\frac{9!}{6!(3)!} =$

84

9.7.5

there are $20^5 = 3200000$ ways to order.

9.8.1

$36^6 - 10^6 - 26^6 =$
1866866560

9.8.2

Total Cards is: 52
Ranks = 13

$$\frac{13!}{5!\,(13-5)!}$$

$$\frac{13!}{5!\,(8)!} = 1287$$

There are 4 suits

$1287 \times 4^5 = 1287 \times 1024$
$= 1317888$

TOTAL number of Combinations Formed as 5 hand Cards using 52 cards : $\frac{52!}{5!(52-5)!}$

$\frac{52!}{5!(47)!} = 2598960$

TOTAL number of Combination with number of Card With SAme Rank

→ 2598960 - 1317888 =

= 1281072

∴ Number of 5 hand Cards With at least two Cards With Same Rank = 1281072

9.8.3

1ST digit = 1 WAY
2nd digit = 1 WAY
3rd digit = 1 WAY
4th digit = 2 WAYS
5th digit = 2 WAYS

6th digit: 2 WAYS
7th digit: 2 WAYS
8th digit: 2 WAYS

$1 \times 1 \times 1 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$

So there are 32, 8-bit strings that begin with 000
TOTAL 8 bit string $= 2^8$

So the number of 8 bit string that do not begin with $000 = 2^8 - 2^5 = 256 - 32 = 224$.

9.8.4

Numbers of ways = N(9 choices for left most positions except VP and arrange rest of 9 persons).

$= 9 \times 9! = 3265920$